In this test, you have to prepare 3 **specifications** with 3-3 **test cases**, the **reduction tables** and small **algorithms** using the specification tool and the structogram tool. Prepare the solutions based on the patterns.
In the postcondition (of the specification), you should use the short form - and the algorithm needs to follow your specification.
Please look after the **good, readable quality** of your work.
There are **90 minutes** available. For a good solution **a maximum of 100 points** could be reached. Don't forget your name and neptun-code! During work you are not allowed to talk (chat), to use cell phones, the internet, books, tutorials, ...
Remember! Cheating is not allowed, and action will be taken.

Name: Jane Doe **Neptun code:** JDJDJD1

1st task

The Librarian of the Unseen University stores the name and the coordinates of each Library in the Discworld. Give the first library in the list, which is further north (the y coordinate is smaller) than the next one.

Example input:
n: 5
Libraries:

name	x	y
L1	12	24
L2	5	15
L3	69	4
L4	12	4
L5	212	47

Output: L4

Specification

```
In: n∈N, pl∈Tpl[1..n], Tpl=(name:S x x:N x y:N)
Aux: exists∈L, ind∈N
Out: lib∈S
Pre: ∀i∈[1..n]: (pl[i].x≥0 and pl[i].y≥0)
Post: (exists,ind)=SEARCH(i=1..n-1,pl[i].y<pl[i+1].y) and
      exists -> lib=pl[ind].name
```

Link of the specification:
https://progalap.elte.hu/specifikacio/?data=H4sIAAAAAAAACs2RwWrCQBCGX2WYk9I1uKm2dGgKUgotiJXqLckhmggL2zUkG4iI0OP2QfpieZKySaRKFeqtOSQZ9pv%2Fn%2F1ni3maLMVKLCMt1goJXxSBqoyZMEhlZcw8lT53HBUymKfS66joPaEZlFDSBErY0KQbZIEaFSVBUopc55UxYwZCxVbFnr0WmkCKRWXMzNbTLCGozIeojGm0CTqp9EXolNXnVx8iFUNTb2xdG0zXuSboNBZMqLjrzZ5Gb4%2FPHeFZjR5nbcu9%2FV7x0Nl0rZJthvppeqH3YIfxLKXi0LEXQoY6yXWO5G8xjnSEhIpgaHtTSeDvRbb19SHAMQ%2BQQUnAXQYbAnewY78ht4WGNcOHp5jrlrm5q6GTOoNjs5PMsGXcPXS7s1BoX1IsCMYD%2B9uEQKCzIrG1UDFBfYIM6ywIkWGW5IXUSHzHDhPhFyVy7B9gP8DDEVaRzH9m6P%2FF%2F5KN%2FP%2BF8LML4WfCCHffye%2FRuLIDAAA%3D

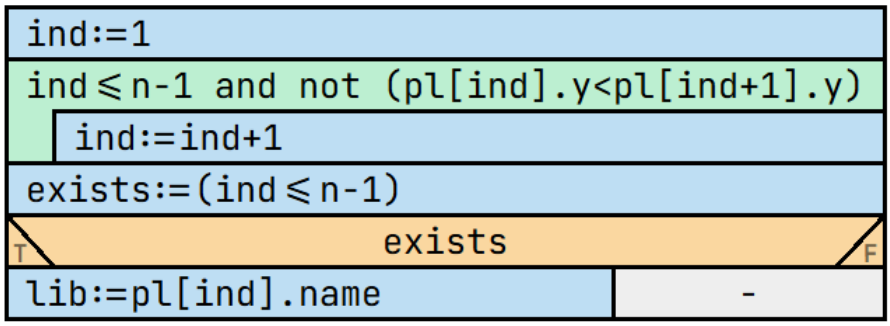
Reduction table:
b..e ~ 1..n-1
A(i) ~ pl[i].y < pl[i+1].y
exists ~ exists
ind ~ ind

Test cases:

<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case
n: 5 pl: [{name: "L1", x: 12, y: 24}, {name: "L2", x: 5, y: 15}, {name: "L3", x: 69, y: 4}, {name: "L4", x: 12, y: 4}, {name: "L5", x: 212, y: 47}] lib: L4 exists: true ind: 4	n: 1 pl: [{name: "L1", x: 12, y: 24}] lib: - exists: false ind: -	n: 5 pl: [{name: "L1", x: 12, y: 2}, {name: "L2", x: 5, y: 15}, {name: "L3", x: 69, y: 4}, {name: "L4", x: 12, y: 4}, {name: "L5", x: 212, y: 47}] lib: L1 exists: true ind: 1

Algorithm



Link of the algorithm:

https://progalap.elte.hu/stuki/?data=H4sIAAAAAAAACq1VYVMaMRD9K84y09Fp6gQQgevwoVqLOIVWbS3q8CGXbCR45vDuENC5%2F97JQUKoOpZp%2BXJhb%2Fe9ty%2BbyxMoAQHQ3Ua1VqbNcpVWq43a%2Fj4QGMYPmHQEBHoSRQRSjJBnKFYRHQtMIXiC0ZkcHU4nXbMu4FyAwJglqDNTBO32yWNvdvvZYCshUEMgWZQigWw%2BRghAx%2FEYcrJKtHhe5QpvIeJlpBTvJ6g5AgE%2BVJFIUHdECsE1fOnIn41LnAKBh3mXnjZHJ0Dg%2BO7TVfRVnQMBnPYOG%2B32FQxyssq2SrzyTTpLM5bhHeoMCGQ4yyAApUXQKpt2x70f95fj%2BNCRuMAaSeMiod879bftc4kWz6v08Lz2NzWRhw%2Fq9vhbrfDI4Vg6D3gTj6ZDFRVcsRYqU7FemPQuyj629IfyFtNiS8fZ1vY4ulZaDHbn5tXiz%2FvyYHe%2BAwTCWMzPl7rXXcvJSrZV6gKbGf3abhZCDJEbJ0vkzdc%2Fjg3OVJqlQWvbs2bHcCraH53d9B8dpwuscfbD9KQ%2FrF28OUUu0eJ5lR6eOy6bT9HlND066vWqxRSdPta7OKIHjs4F1uh%2B9eezUTfK3pTvEi2eV%2Fl%2F5DuBRr7DsXQe8CY7ruQfJ2Cx32b%2FkwkeJEzz4dp8e7tSIL2Q4jrPycpzq9MFNpuSFyYzUmHQsmdTszuEPCeQZsmEZ%2FFNwu6Ma09QvAmgy5Q232CWKBZG5h4BIJDE8TOj0mE8vVilLXWY6DEygckylA%2BMcVGcFFfSQtM1lKSklFIgUDJPaewtScldLKQ1t5LFqk6rlBUrjkw26jBYtHjA%2BO1NEk%2B0WAKHlbBi0szTAiO3MaxzZleyWIk9TrFi4MzN%2BQyOc84LVs65hZMVG0Mp%2BGtwnsfPUM1v2fNf4UvZbC7eUtrgTfNpLCHintj3Hcnz362Fd%2F05CAAA

2nd task

Yossarian marked on a square grid map (n x m) how many times he had been to each place. Which place had he been to the most? Give the "coordinates" of the first such place.

Example input:
n: 6, m: 4

12	3	1	4
3	2	23	5
4	2	2	1
5	2	23	0
0	2	4	1
0	0	0	1

Output: (2; 3)
(2nd row, 3rd column)

Specification

```
In: n∈N, m∈N, map∈N[1..n,1..m]
Out: mpx∈N, mpy∈N
Fn: maxcol: N->N x N, maxcol(i)=MAX(j=1..m,map[i,j])
Pre: ∀i∈[1..n]: (∀j∈[1..m]:(map[i,j]≥0))
Post: (mpx,)=MAX(i=1..n,maxcol(i).maxval) and
      mpy=maxcol(mpx).maxind
```

Link of the specification:
https://progalap.elte.hu/specifikacio/?data=H4sIAAAAAAAACsVQPWvDMBD9K8dNNlyNZTsdDlzoUujQtGPB8SASFxQiRcROUQiBjO4P6R%2FzLymy3ZR0SNup0nCnp%2FfefeyxttVcvai5bNTaIOO9YTBd204J9Bik9UkhosiQiCJdzjYz87htGLR1I8fufOI%2F7gyDlm6%2BXjFMr26m4KA38Uigwvzh9jlY5t6HtLSFomUZet3TpmLo2qPq2ravVTIEXXtcjm9dcvAp6N7e43BQreuGIdDW0WCt8r7PU71IS%2FcqVyFIs%2FAC6I%2B2u3ykaOt6kjILJGyquqmRiz0uZCOR0TBce51myPooLUNRiIRSEpSV9GVapJRQktLkDMw8SOIMmwzE%2BAyMKaHsGzEmf0XZr1xbx5AM2Y4h9RkSGqkrZETCTVVvVw2yONCP%2Ff%2FbAOI0gPhN8%2BLS8v%2FiWh4%2BANqnw%2BftAgAA

Reduction table:
MAX in the rows:

maxind , maxval	~	mpx, -
e..u	~	1..n
f(i)	~	maxcol(i).maxval

MAX in the columns (maxcol)

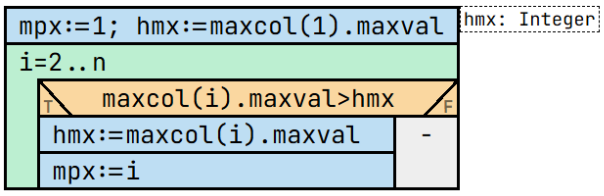
maxind , maxval	~	mpy, -
e..u	~	1..m
f(j)	~	map[i,j]

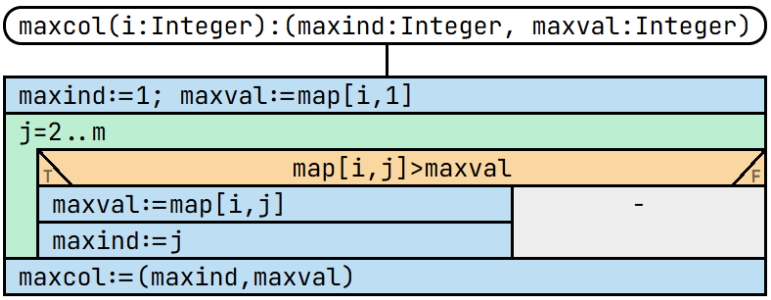
Test cases:
<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case
n: 6 m: 4 map: [[112,3,1,4], [3,2,23,5], [4,2,2,1], [5,2,23,0],	n: 6 m: 4 map: [[12,3,1,4], [3,2,23,5], [4,2,2,1], [5,2,23,0],	n: 1 m: 4 map: [[12,3,1,4]] mpx: 1 mpy: 1

[0,2,4,1], [0,0,0,1]] mpx: 1 mpy: 1	[0,2,4,1], [0,0,0,1]] mpx: 2 mpy: 3	

Algorithm





Link of the algorithm:

https://progalap.elte.hu/stuki/?data=H4sIAAAAAAAACq1X227aShT9lWpHOmqkUWTM3VUewlGToLaEilwOiXgYZsaXYI%2FBGAKN%2FO9HMzDbQwNKUcMLo82etda%2B2rxCxMED56xVrVecdqXqVKuteqMBBMJ0KbIuB08u4pjAXMSC5YKXFplyMQfvFSZ3ThBeLO7UWcOhgcCUZkLm5aUw4lxI8HwazwWBfD0V4MFczBZCMgEEWBjFPBOyy%2BfgPUHAkt7gdh4AgWqz2e%2F%2Bu2jBqCCl3XBajiWnreQAc05zkQiZA4FcrHLwIJmuvPPKl09hsvLOE7piafy5cnqW0NWSxlAQ%2BO5euL%2BW%2BTVyo2GHG%2FUeH3U9mTUur9YrHSriGDoL%2BJhQX8Io1lyp5FEepRI8iM7dszMJBMYpXw%2B2ero7MRUEfnYmg%2Fnq8QlVoGFHAao%2BPuAef5wN0%2FA7EGjX2kE6vP%2BqY2%2FkV994%2B2GCzGjYjb2fX8rZbf8gs0zTqYoEHbFVy5sfEwkKVPIRx9BZwBad1T%2F76SL%2Ft7pt2zIybflPkH8JEwWbZwvRyahk4U45rYpp6D0umIqClPUwwq0CWcIt1D8eL3usInussPCGEw1%2FzalHOlIkV50btv46rCAJGnZIms1OT97w7rsNhY4Gz7r5t7uPcZdLEanevH3JqdPtRUBgfT0eTLkvdIOhi6G37hwTzr6c0VUkud6Emxqpqk2fIlIZqcD5A3dl1hwjMxp2mNfDx7a7bB6eJJNIdDR41k0Lz0rEsem8XLu9ycNkoROHOIbOAj4mcfuW6rNaqsm%2BpYoxFQRq60heyXb5FEHD7lNk8Fh3JtNv7yYQHfExUd608DALxydQDNNf4%2F7PDAjU7mrBj8nFUOfyvh%2F892Malk2Iht1WmN0MI8pe3m8F44itUN78mEhQoJKPOIbOAj6mk%2FcsaTUuzyO1nLdr7uB%2Btop1aD9jFgpSlsJotmpzTPPsn3tr2J%2F1sGO5sVeN4SPo9Jp51pU3uw0rbwwfsc5YGnvnnzeEZBPmKRQFgXmeLVieBhlNVHu8gqSJggECS5pFdByrF1395PrUlbkIhMpzlqZv3rzmYfpyLSgXGUpSpvsSRdW%2FIEhhHoTeFvfU2wo0BmK2r3H4TZQtxMqLLURRvtGh1RUj1bFxmun3%2BE2mnuDE9x3HcYDAifr2VV%2Bf%2BD5D29ip48nXp6ZTdag%2BMUH9VhNGm8R3KJsEWbqQfAs8dseuclPfBlgwYxNNRs3J1ydeY45wFZz6u%2FEGjjHGNCtjzMD5rrEJn7NDcFbN36CqzzbmP8L3%2FXZ786vjtFi7rk5CiBpv2Bkpiv8BaY1yZG4NAAA%3D

3rd task

For N days, Trurl recorded how many robots he built. List the days on which he built more robots than the total number of robots built in the previous D days (1<D<N)??

Example input:
n: 10, d: 3
2, 4, 5, 1, 13, 1, 1, 1, 4, 5

Output: 5, 9

Specification

```
In: n∈N, d∈N, tr∈N[1..n]
Out: cnt∈N, days∈N[1..cnt]
Fn: total: N->N, total(i)=SUM(j=i-d..i-1,tr[j])
Pre: n>d>0
Post: (cnt,days)=MIS(i=d+1..n,total(i)<tr[i],i)
```

Link of the specification:
https://progalap.elte.hu/specifikacio/?data=H4sIAAAAAAAACo2PwWrDMAyGX0XolDAl1MnCqFhyHPTQblB2SnMwSQYunTts9VBK733OPcmw17JcNoaxpR9J32%2Bd0H%2BMvXkzvRazt8i4sAz283JZEQzfQVyIrcpz223cxj4fhKG3cm3SR3%2Br91Zix5NlkL3oHcMqawIiiMSk9fp1mWxrkw15bjJF4tptl4aRFzcy2GZoZlHtvTAkvRUK%2FLReLtaJqYe78Am60R7FtaYjkyKhjF48cnvCQYtGRsugImtgKCEk4hjagu6pIkWqDA%2BpILtY7q0wFHFCHz1DW9E8bhMuElr9PiIjErrRH3aCrM40tZv%2F4XY9xcRqNrHq%2FsOvrvzih68oODyQmoLLCbiM%2B%2F1C785fH2YgwP8BAAA%3D

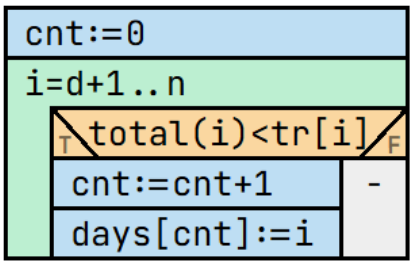
Reduction table:
MIS
b..e ~ d+1..n
f(i) ~ i
A(i) ~ total(i)<tr[i]
cnt, y ~ cnt, days

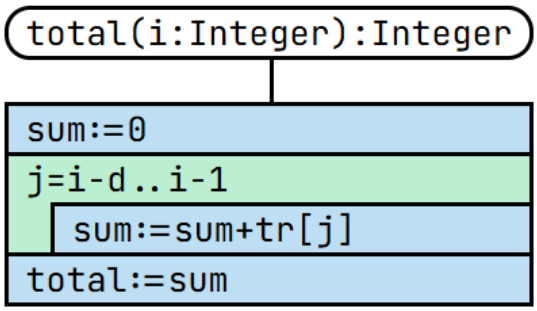
SUM
b..e ~ i-d..i-1
s ~ total()
i ~ j
f(j) ~ tr[j]

Test cases:
<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case
n: 10 d: 3 tr: [2,4,5,1,13,1,1,1,4,5] cnt: 2 days: [5,9]	n: 9 d: 3 tr: [2,4,5,1,1,1,1,1,2] cnt: 0 days: []	n: 5 d: 2 tr: [1,2,4,7,12] cnt: 3 days: [3,4,5]

Algorithm





Link of the algorithm:

https://progalap.elte.hu/stuki/?data=H4sIAAAAAAAACq1WXW%2FaMBT9Lxdp2lQPha8FMvWFqtPQyrauGuuEeDC2A26DDY6h66r898kpvjErbEMdL7EO9jnnnmvd5AEkhwSieqvbbrUavXbc7rQ6b7rNDhCY640wAw6JWmcZgVxkglnBK0RpLnJIHiCdp8NufP%2FFrUtCBAgsqRHKVofmknOhIElplgsC9n4pIIFcrNZCMQEE2Fxm3Ag14DkkY%2Fh4cbXJ5xfXQKBD9cVINjKYFKTCvWawsdIMnRxQttSKhVAWCFjxw0ICTNnkNIKCgMw3i8HqQxNFENgRQWPHl9ccfh69G242ZU3I4%2BUC4mNqupvLrNTSiksrtXLOT%2FlJo15XQGCq%2Bf3V1tFgp6qCwBmdbj6dCYMeENjx8HWkr9vvlvroXJmyJw0nhAxeKKAMhDCg47MNnJ%2Fbsxt2x2dlzA3V7Dd%2BXkpURmBHuf3p%2Bv1td3V7UFlpvXSV4EbPF5z8P5WgQWcfebxcQBzIBVd1v5xMf7siVluavZSvXmT2rTVjOXGtM2vRN1Sx%2Bc6NCXpVku7ZgiEUQfzeMgLPvlSc3udjpuwkOZVOynRu6ejc3KEUAjtS52wgVn0%2B%2FGtzcSNar04%2Bd7aJYa8tp80YCHzN%2B996MV8DgUGjs4qWlzdls3GLlw%2FOHFPOnuTy9WI75i6N6tPllUIRBHZE4vj7qrW5Ojx3fGa40fMFJ8N2VzUfm9zsw%2BKnZIvzMiPkwUlSER%2BT0b6xeXMqX%2FN6Xb5u7BucWFdBKkveBQLHhXioU%2Fl6cWLN%2BGbixPCOeDEEnn0tyilQykFREMitWTOrZ4YuXPIPoOjCnRxS6a7HhhpJp5n7DgAgYLR%2B8o7K5%2FpuVG3bGnDoe0G5e9OUUEGQezuIkoGyYibMK7%2F4g15Qacjsxtd%2BB8XENTrTpvyCeax9DLU0jaIoAgI190zdiKylKUNsGnVwlZarOGpFtFwxQdNuDJPHKPuU3c6MXiu%2BJZ42p24Y19zTEwvmMREz6ldpueJtFommo3MfWk%2FoGGOsVGWMebq06TGRcnaILmjpE1b329b8T%2Fxp2us9%2FhtFXdZz3401IUSbvwkTKYpfvSItpmoKAAA%3D